Project Overview

Project Details

Intern Name: Muhammad Talha

Internship Program: Cyberster Blue Team Internship

Organization: Cyberster

Week 4 (Days 22–28)

Date: July 16, 2026

Status: Completed

Executive Overview

During Week 4 of the Cyberster Blue Team Internship, the security laboratory was significantly enhanced to better simulate a real-world Security Operations Center (SOC). The primary focus of this week's activities was to improve network visibility, strengthen monitoring capabilities, identify potential system vulnerabilities, and present security information through centralized dashboards and professional reporting.

Several defensive technologies were successfully implemented and integrated into the existing laboratory environment. These improvements enabled more effective monitoring of network traffic, better correlation of security events, and enhanced visibility into the overall security posture of the environment.

Unlike previous weeks, which focused mainly on event collection and detection, Week 4 emphasized proactive security management by introducing firewall protection, threat intelligence integration, vulnerability assessment, and executive-level reporting.

Key Achievements

One of the major accomplishments this week was the successful deployment of **pfSense** as the primary firewall for the virtual lab environment. The firewall was configured to manage traffic between all virtual machines, creating a controlled network boundary similar to what is commonly found in enterprise environments.

Firewall logging was enabled, allowing network events to be collected and forwarded to the Wazuh platform. This centralized approach improved visibility into network communications and simplified the investigation of security-related events.

Threat intelligence capabilities were also explored to understand how external security information can provide additional context for detected activities. Rather than relying solely on raw alerts, the environment became capable of enriching events with publicly available intelligence sources, helping analysts determine whether observed indicators are associated with known malicious activity.

Another important milestone was enabling vulnerability detection within the Wazuh platform. The environment was assessed for outdated software and known security weaknesses. The results demonstrated how proactive vulnerability management can help reduce organizational risk before vulnerabilities are exploited by attackers.

Finally, a customized security dashboard was developed to present important security metrics in a clear and organized manner. This dashboard provides a centralized overview of system health, alert activity, firewall events, and vulnerability information, allowing security teams to quickly assess the overall condition of the environment.

Security Assessment

Throughout the reporting period, no evidence of an actual security compromise was observed. The alerts generated during the week were intentionally produced through controlled laboratory exercises designed to validate the monitoring and detection capabilities of the environment.

The overall security posture of the lab remained stable throughout testing. The implementation of pfSense improved traffic control, centralized logging increased operational visibility, and vulnerability detection provided valuable insight into software risks that require future attention.

These enhancements collectively strengthened the defensive capabilities of the laboratory and provided a more realistic representation of how modern Security Operations Centers monitor and protect organizational assets.

Business Impact

Although the activities performed during this week were conducted within a laboratory environment, they closely reflect security practices implemented by modern organizations.

The introduction of centralized firewall monitoring improves visibility into network communications and supports faster identification of suspicious activity.

Vulnerability management enables organizations to identify weaknesses before they are exploited, reducing the likelihood of successful cyberattacks.

Threat intelligence integration provides additional context that helps security analysts prioritize alerts more effectively and focus on the most significant risks.

Professional dashboards and reporting improve communication between technical teams and management by presenting complex security information in an understandable format.

Together, these capabilities contribute to stronger security governance, improved incident response, and more informed decision-making.

Recommendations

Based on the activities completed during Week 4, the following recommendations are proposed:

- Continue monitoring firewall activity to identify unusual network behavior.
- Review vulnerability findings regularly and prioritize remediation based on business risk.
- Keep operating systems and software packages updated with the latest security patches.
- Maintain threat intelligence integrations to improve alert accuracy and investigation quality.
- Review dashboard metrics daily to detect emerging trends and abnormal activity.
- Perform periodic firewall rule reviews to ensure only required network communications are permitted.
- Continue expanding security monitoring by integrating additional defensive technologies and log sources.

Overall Risk Rating

Following the successful implementation of the planned activities, the overall security posture of the laboratory is assessed as **Low to Moderate Risk**.

No critical incidents were identified during the reporting period. The majority of recorded events were generated as part of controlled security testing conducted to validate monitoring and detection capabilities. The implemented security controls significantly improved visibility across the environment and established a stronger foundation for future defensive operations.

Conclusion

Week 4 marked an important milestone in the Cyberster Blue Team Internship by transforming the existing laboratory into a more complete and realistic Security Operations Center environment.

The deployment of pfSense, integration of centralized logging, implementation of threat intelligence, activation of vulnerability detection, and development of a customized monitoring dashboard collectively enhanced the laboratory's defensive capabilities. These improvements demonstrated how multiple security technologies work together to provide continuous monitoring, proactive risk identification, and effective security reporting.

The experience gained throughout this week strengthened both technical understanding and analytical skills while emphasizing the importance of clear communication between security teams and organizational leadership. The completed environment now provides a solid foundation for more advanced security operations and future incident response activities.